

Paper 8: Two Accountings — Condensation, Internal Expansion, and the Area Law

Inversion of the Splitting Theorem under Shared-Curvature Accounting, the Jeans-Type Threshold, the Internal Clock and $a \propto t^{1/2}$, and the Vacuum as a Reservoir

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Abstract

The no-exception splitting theorem of Paper 5 was based on **local accounting**, in which each fragment carries its own container, and its conclusion was complete dispersal. The core of the present model, however, is that $R^2 = \sum \nu^2$ sets a curvature-like scale, and local accounting on a flat background is not consistent with the model's own claim. This paper puts the curvature into the accounting and obtains three results.

First, the **inversion of the splitting theorem**: when all fragments ride on a single shared container determined self-consistently by the total content S , minimizing the unfilled amount becomes equivalent to maximizing the total capacity, and an exhaustive check for $S \leq 25$ shows that **condensation (the single fragment) is strictly optimal**. Local accounting (surface term $s^{3/2}$, subadditive) $\rightarrow$ dispersal; shared accounting (capacity term s^2 , superadditive) $\rightarrow$ condensation — the divide is a single exponent, and physically it is the choice of whether the container = the curvature belongs to each fragment or is shared. In a one-parameter family mixing the two accountings, the condensation onset threshold behaves as $w^*(S) \approx 3.40/\sqrt{S} \rightarrow 0$: **the larger the system, the more a slight sharing of geometry suffices for condensation to dominate** (a scaling of the same form as the Jeans criterion; a structural correspondence, not an identification). Under the exclusion rule (Paper 7), the competitor on the dispersal side changes from the all-1 partition to a Fermi-sea-type configuration, so the threshold drops further ($0.6397 \rightarrow 0.5872$ at $S = 9$).

Second, the **internal expansion law**: when the internal splitting cascade of a closed state space (container fixed, no exterior) is read through the ratios of an internal observer (who uses their own fragment as the measuring rod and carries the dual clock $\tau = 1/R'$), the apparent scale factor satisfies the exact identity

$a^4 f = \pi^2/2$ (f is the occupancy fraction) and, in the leading term, follows

$$a \propto t^{1/2}$$

(a universal law independent of the branching number; the fitted exponent in the central window of a deep ternary cascade is 0.507). $f \propto a^{-4}$ is the 4-dimensional dilution of a conserved number, and the appearance of a radiation-type dilution law in a system whose content is oscillation only is a consistency check. Expansion requires neither an exterior nor growth of the container: **a system that subdivides inward looks expanding when measured with a measuring rod that itself shrinks**. We give the numbers including the exclusion-rule correction of the endpoint (the cascade terminates at $s' = 3$, and the all-1 state is unreachable).

Third, the **area law and the reservoir**: the phase-space area per state is a rigid body of exactly 2 ($= 4\delta_{\min}^2$), and the conservation of total area is the same law as the dual reading of energy conservation. On the other hand, imposing exact conservation of mode area freezes the cascade at stage 0. The exact conservation law that actually holds is “occupied area + gap = container volume,” and splitting is the process of transferring occupied area into the gap. **The vacuum-like state (the gap) is not a structureless void but a reservoir of area that makes evolution possible.**

Keywords

condensation, dispersal, shared container, Jeans-type threshold, internal observer, scale factor, radiation-type dilution, area law, rigid tiling, reservoir, occupancy fraction

1. Introduction

The classical insight into self-aggregation under gravity goes back to the Jeans criterion [5], by which the instability of a uniform medium depends on scale (the larger the scale, the more gravity dominates). Support by degeneracy pressure (Chandrasekhar [6]) is the classical example of a structure in which exclusion statistics opposes aggregation. The viewpoint that the accountings of geometry, thermodynamics, and information are two sides of one ledger is represented by Jacobson’s thermodynamic derivation of spacetime [7].

This paper does not extend these physical theories. What it does is to establish, **inside the counting model**, by exhaustive checks and exact identities, the following three points: (i) the choice of accounting (local or shared) switches between dispersal and condensation; (ii) an expansion-like power law appears in

the reading of an internal observer; and (iii) the area ledger requires vacuum-like regions as a reservoir. All correspondences with standard cosmology (the scale factor; the power law of the radiation-dominated era [8]) are recorded as structural correspondences with alarms attached, and no identification is made.

2. Two Accountings and the Inversion of the Splitting Theorem

2.1 Local accounting (review)

The splitting theorem of Paper 5 [2] is based on the local accounting $\Delta_{\text{loc}} = \sum_a [V_4(R_a) - N_0(R_a)]$, in which each fragment a carries its own circumscribing container $V_4(R_a)$. The leading term of the volume gap is the surface term $\propto s^{3/2}$, and by the subadditivity associated with the power $3/2 > 1$, consistent splitting reduces the gap without exception — the conclusion is complete dispersal.

2.2 The shared container (global accounting)

In the present model (Paper 4 [1]), $R^2 = \sum \nu^2$ sets a curvature-like scale (it can be written in the form $\Lambda_{\text{model}} = 3/R_c^2$). We therefore let all fragments ride on a single container $S^4(R_c)$, $R_c^2 = S$, determined self-consistently by the total content $S = \sum_a s_a$. Under fixed S the container volume is the same for every partition, so minimizing the global mismatch is equivalent to **maximizing the total capacity** $\sum_a N_0(\sqrt{s_a})$ (by this equivalence the conclusion does not depend on the choice of the container constant).

Results of the exhaustive check of consistent partitions for $S \leq 25$:

1. **Odd S :** the single fragment (S) is, without exception, the unique optimum.
2. **Even S :** since no single slot exists, the optimum is always $(S-1, 1)$ — the largest fragment plus the minimal satellite. Not the symmetric partition but the **maximally asymmetric** one wins (an accretion-type “rich get richer” structure due to the convexity of the capacity).

Figure 1. Shared-container accounting (exhaustive for $S \leq 25$, with the capacity law included): the capacity of the single fragment strictly exceeds that of every partition.

2.3 Restriction of merger channels by the Z_2 selection rule

Since consistent fragments carry odd labels, odd + odd = even has no parent slot, and **two-body mergers are forbidden**. The minimal allowed merger, once the exclusion rule (Paper 7 [4], $c(1) = 1$) is included, is $(3, 3, 1) \rightarrow 7$.

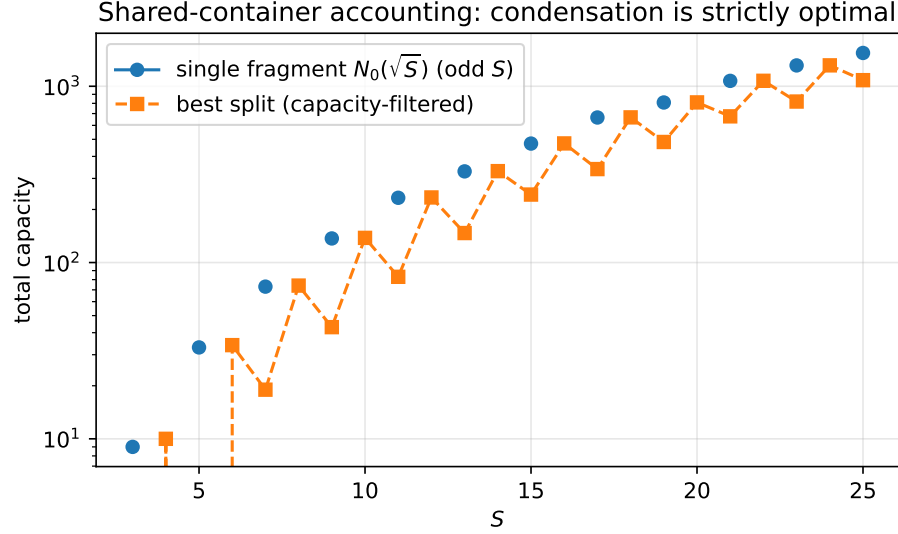


Figure 1: Figure 1. Condensation optimality under the shared container

2.4 The Jeans-type threshold

By an exhaustive check of a one-parameter family in which the container is mixed as “local $(1 - w)$ + shared w ”, the condensation onset threshold behaves as

$$w^*(S) \approx \frac{32}{3\pi} S^{-1/2} \approx \frac{3.40}{\sqrt{S}} \rightarrow 0$$

(the exact formula $w^* = 1 - (N_0(\sqrt{S}) - S)/[\frac{\pi^2}{2}S(S-1)]$ has been checked against bisection). **The larger the scale, the more the condensation side wins** — a scaling of the same form as the Jeans criterion [5].

Figure 2. The Jeans-type threshold $w^*(S)$: the exact formula and the asymptotic form $3.40/\sqrt{S}$. The star marks the revised value 0.5872 at $S = 9$ under the exclusion rule.

2.5 Revision by the exclusion rule

Under the capacity law of Paper 7, the maximal dispersal within a single lattice is not the all-1 partition but the **Fermi-sea type** (occupying one cell at a time from the lowest shell, $n_{\max} \approx \frac{\pi^2}{2} M_F^2 \sim S^{2/3}$, $M_F \approx (3S/\pi^2)^{1/3}$). Since the competitor on the dispersal side is weakened, the condensation threshold drops: at $S = 9$, $w^* = 0.6397 \rightarrow 0.5872$. It is worth noting that exclusion statistics

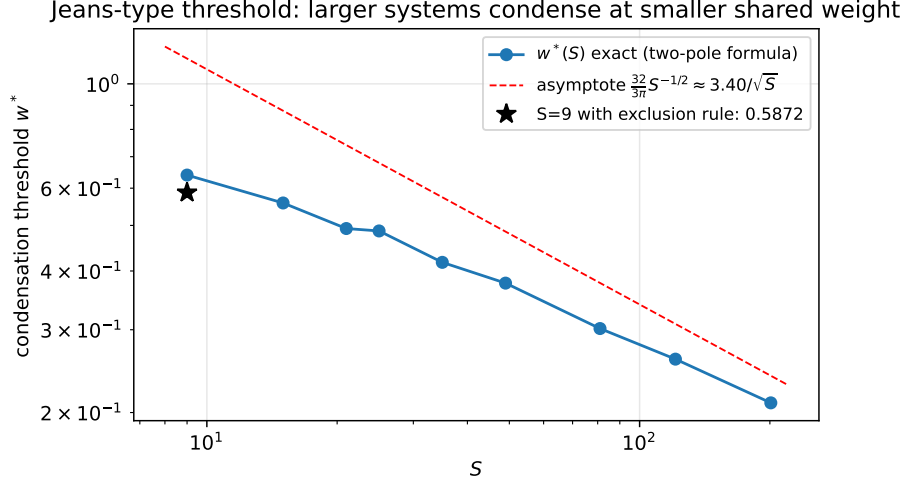


Figure 2: Figure 2. Jeans-type condensation threshold

appears on the dispersal side as a degeneracy-pressure-type structure (of the same form as Chandrasekhar [6]; a structural correspondence).

2.6 Two attractors of the occupancy fraction

On the self-consistent container, the occupancy fraction is $f_{\text{disp}} \approx 0.0848 S^{-4/3} \propto \Lambda_{\text{model}}^{4/3}$ in the dispersed state (Fermi sea, the revised value including the exclusion rule) and $f_{\text{cond}} \rightarrow 3/16$ (a universal value) in the condensed state. The two-phase coexistence of a dilute global background and packed local clumps is recorded as a qualitative observation (the mapping to observables is quarantined in the appendix with alarms attached).

3. The Internal Expansion Law: $a \propto t^{1/2}$

3.1 Setup

Consider the internal splitting cascade of a closed state space ($R_c^2 = S$ fixed, no exterior). When the fragments carry a representative value s' and there are $n = S/s'$ of them, defining the apparent scale factor as the ratio of the mean inter-fragment distance to the fragment size, $a = d/R'$, it satisfies with the occupancy fraction the exact identity, in the leading term,

$$a^4 f = \frac{\pi^2}{2}.$$

Apparent expansion and rarefaction are two readings of one and the same quantity, and $f \propto a^{-4}$ has the same form as the dilution of a conserved number in 4 dimensions (radiation type).

3.2 The internal clock and the main result

The proper period of each fragment is $\tau = 1/R' = \lambda'$ — **the dual relation $\nu\lambda = 1$ is itself the clock** (Paper 6 [3]). The Z_2 selection rule forbids binary trees, so the minimal cascade is the ternary tree. For a k -ary self-similar cascade:

- With respect to the stage number: the e-folds per stage are $\ln k/4$ (exponential growth).
- **With respect to internal time:** from $a^4 \propto 1/s'$ and $t \propto 1/\sqrt{s'}$, in the leading term

$$a \propto t^{1/2}$$

independent of the branching number k (it holds for an arbitrary geometric schedule). The fitted exponent in the central window of a deep ternary cascade ($S = 3^{14}$) is 0.507.

3.3 The radiation law as a consistency check

$f \propto a^{-4}$ is the dilution law of radiation, and the content of this model is ν (oscillation) only. **That the radiation-era law appears is the correct answer; a matter-era or constant-term-era law would have been a contradiction.** This is not a fit but a consistency check (the correspondence with cosmology is a structural correspondence [8]).

3.4 The endpoint correction by the exclusion rule and the specificity of the clock

By the capacity law the final stage $3 \rightarrow (1, 1, 1)$ is forbidden, and **the cascade terminates at $s' = 3$** (the total number of e-folds is $\frac{1}{4} \ln(S/3) + \frac{1}{4} \ln(\pi^2/2)$). The endpoint correction factor remains, by the coincidence $N_0(\sqrt{3}) = 9 = 3^2$, exactly $\pi^2/2$, unchanged.

We also checked robustness with respect to the choice of clock: $a \propto t^{1/2}$ is a law **specific to the dual clock**. With the area clock (cumulative released area), time saturates at a ceiling and no power law appears; with the stage clock, the law is exponential. The precise status is: “the radiation-era law when $\nu\lambda = 1$ is chosen as the clock.”

Figure 3. (a) The internal expansion law $a \propto t^{1/2}$ (exact N_0 , termination at $s' = 3$). (b) The exact conservation law: occupied area + gap = container volume (the vacuum = the reservoir).

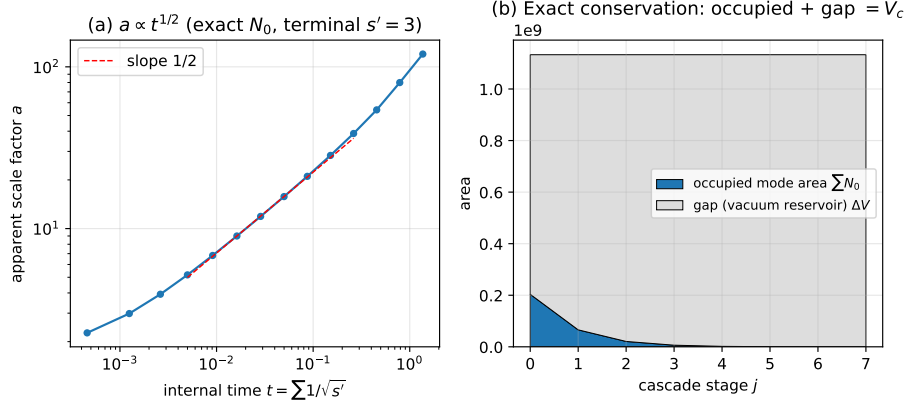


Figure 3: Figure 3. Internal expansion law and the vacuum reservoir

3.5 Interpretive implications (within the model, restricted)

- (i) Expansion requires neither an exterior nor growth of the container — expansion is another name for the subdivision of the measuring rod. (ii) Regions that appear mutually distant are, when the hierarchy is traced back, internal subdivisions of the same parent cell, so no separate assumption of initial causal contact is required (an observation within the model; it does not claim to solve the horizon problem). (iii) The “beginning” can be read, without any explosion, as the onset of the splitting cascade from an undifferentiated single cell of the highest frequency.

4. The Area Law and the Vacuum as a Reservoir

4.1 Rigid tiling

The jumps of $N_0(\sqrt{s})$ occur only at odd s , and each consistent label m owns a window $[m, m+2)$ of width exactly 2 on the s axis. The area per state,

$$\Delta s = 2 = 4\delta_{\min}^2,$$

is a **rigid body** (upper bound = lower bound), and the conservation of total area $\sum \delta_a^2 = S/2$ is the same law as the dual reading of energy conservation $\sum \nu_a^2 = S$. The logarithmic area and the (ν, λ) area agree exactly on the constraint (Jacobian $1/\nu\lambda = 1$), and the necessary and sufficient condition for splitting to be realizable as an area-preserving map (piecewise symplectic) is the summability of the areas (a Moser-type existence theorem).

4.2 Strong conservation freezes evolution

If area is read as the mode number $\sum N_0$ and exact conservation without a reservoir is imposed, then in every consistent partition with $s \leq 25$ the mode area decreases without exception (the strict superadditivity of N_0). Hence **under an area conservation law without a reservoir, the cascade freezes at stage 0.**

4.3 The true conservation law and the reservoir

The exact conservation law that actually holds is

$$\text{occupied area } \sum N_0 + \text{gap } \Delta V = V_c = \text{constant}$$

(verified to hold identically at every stage of the ternary cascade). Splitting is the process of transferring occupied mode area into the gap, and the release is front-loaded (the first splitting releases the fraction $1 - 1/k$ of the capacity).

The vacuum-like state (the gap) is not a structureless void but a reservoir of area that makes evolution possible. Condensation (§2) can be read as the withdrawal of area from this reservoir (the reverse process).

5. Scope of Claims of This Paper

What this paper claims: (1) The optimality of condensation under shared accounting (exhaustive check) and the local/shared divide by a single exponent. (2) The restriction of merger channels by Z_2 and the exclusion rule. (3) The Jeans-type threshold $w^* \propto S^{-1/2}$ (with the exact formula). (4) The identity $a^4 f = \pi^2/2$ and $a \propto t^{1/2}$ (specific to the dual clock, independent of the branching number, with the endpoint correction included). (5) Rigid tiling; area conservation = the dual reading of energy conservation; occupied + gap = constant; and the reservoir structure.

What this paper does not claim: (1) A derivation of gravity, cosmic expansion, or vacuum energy (“condensation,” “expansion,” and “vacuum” are descriptions of configuration comparisons and counting ratios within the model). (2) A force as a function of distance (what this paper exhibits is a configuration preference, not a force; for the quantification of position on the λ side, the relational geometry of Paper 7 is the first step). (3) A within-model derivation of the mixing parameter w . (4) An explanation of the numbers of observational cosmology (the mapping in the appendix is a record with alarms attached and has no independent support).

6. Conclusion

A single choice of accounting — whether the container = the curvature belongs to each fragment or is shared — switches between dispersal and condensation. Shared curvature gives condensation; internal splitting gives the expansion-like power law $a \propto t^{1/2}$; and the area ledger gives the vacuum as a reservoir — each with no additional mechanism. The model has both the dispersing tendency and the condensing tendency built in, and the divide of their competition is quantified as a scale-dependent threshold.

The central observation of this paper is that the three aspects — “the coexistence of a dilute global background and condensed local clumps,” “the radiation-type dilution law,” and “the vacuum as the fuel of evolution” — emerge from nothing but the inventory of $\nu\lambda = 1$, the zeros, and the asymmetric one bit.

Appendix A: Mapping to Observables (a Record with Alarms Attached)

Substituting the observed occupancy fraction $f_{\text{obs}} \sim 1.6 \times 10^{-31}$ (the nondimensionalization of 0.25 hydrogen atoms per m^3) into the identity for the dispersed state gives $S \approx 2 \times 10^{22}$, and the scale width of a single hierarchy level is about 11 orders of magnitude, which does not reach the observed scale width (36–61 orders of magnitude). **The single-hierarchy mapping is rejected** (recorded as a sound negative result). The cross-check for the case where nesting of hierarchy levels multiplies the widths has not been carried out, and the recovery of testability requires an independent third dimensionless observable.

Appendix B: Essentials of the Reproduction Procedure

By the full enumeration of consistent partitions ($S \leq 25$), bisection for the mixed accounting (40 iterations), the cascade tables ($k = 3, m = 14$, exact N_0), the tiling check ($s \leq 40$), and the verification of the reservoir identity at every stage. The scripts and research notes are provided in the public repository (<https://github.com/WurabeSeiji/ai-chat-logs-open>).

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